Bern Match

From a company's need to a verified shortlist of who to contact in Bern. Powered by Apertus.

Technical report · Apertus Hackathon, challenge of the Bern Economic Development Agency · 19 October 2026 · Dimitry, Maxim, Ruihan · Live: kaktamsdengami-bern-match.hf.space · Code: github.com/AnAsteria/Apertus_Hack · All numbers are measured; sources are the repository's `docs/MEASUREMENTS.md`, `docs/report/` and the linked issues.

0.82
Recall@10 on the frozen evaluation set after Apertus refinement (0.77 instant, n = 17)

100%
of cards link to their source page; quotes byte-identical to the stored sentence

< 1 s
first results, no LLM on the fast path (server p95 ≈ 0.5 s)

0 tokens
per plain search; ≈ 2.9k Apertus tokens per refined request

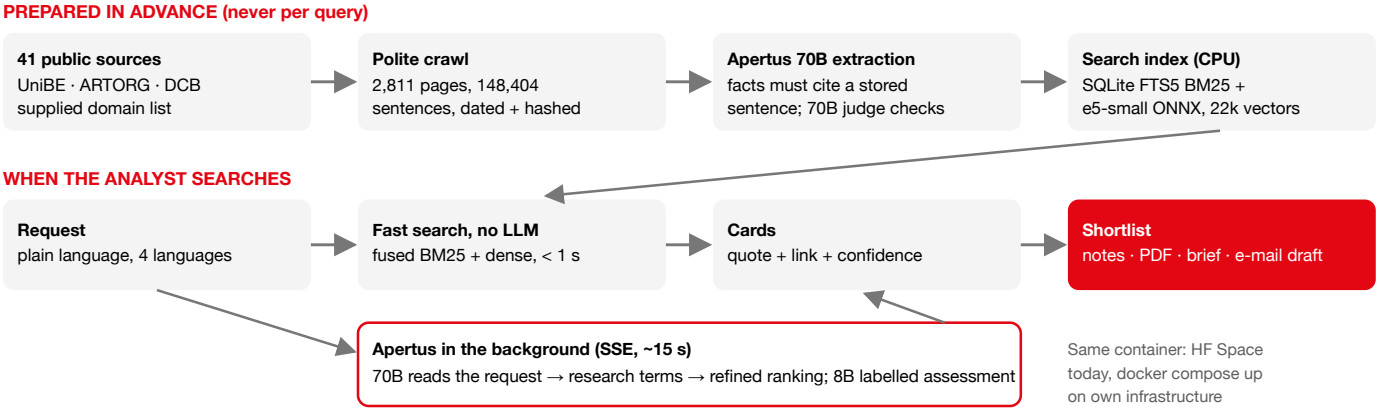
Challenge success criteria at a glance

Criterion	Status	Evidence (frozen evaluation split unless noted)
1. ≥ 80% of expected experts/units in the top 10	**Met after refinement**	Recall@10 **0.82** with Apertus refinement; 0.77 instant (n = 17); development 0.82
2. Different wording (synonym, lay, other language)	**Partly**	Synonym 2/2 · cross-lingual 0.67 → 0.78 with Apertus terms (n = 9) · lay 0.5 (n = 2); development: cross-lingual 0.82 (n = 11), lay 1.0 (n = 2)
3. Every result traceable, nothing invented	**Met**	100% of cards link to the source; 0 quote mismatches; facts kept only if a cited sentence states them; semantic support is weaker on ranks 4–10 (section 4)
4. Structured fields where the source provides them	**Partly**	Person, role, unit, expertise, projects, infrastructure/services, reason for match and source link are rendered when supported; 26% of cards carry at least one structured field
5. All three sources	**Met**	UniBE (inventory-based), ARTORG, DCB: 41 domains crawled; 34 appear in served results
6. Easy to extend	**Met**	A new website is one YAML file validated by the loader, then an incremental build (example: BFH; `docs/source-extension.md`)
Ethics: say when information is insufficient	**Partly**	Answer labels (*low confidence* / *no evidence* / *too broad*); Apertus scope check refuses nonsense; the 2 "no answer" questions still get low-confidence cards

1. Problem and product

The Bern Economic Development Agency connects companies with research partners. An analyst today reads dozens of institute websites in German, English and French to find the right group, and has little to show as proof. **Bern Match turns the company's need, written in plain language (EN/DE/FR/IT), into a shortlist of contacts**: research groups, labs, platforms, projects and people at the University of Bern, ARTORG and the Diabetes Center Berne. Every card quotes the exact sentence from the source page with its link. The analyst collects entries across several searches into a **shortlist**, adds notes and exports a PDF for the team. Optional Apertus drafts (one-page brief, e-mail) are labelled as AI and are never sent by the system.

2. Architecture



Offline. A bounded crawl stores page versions and sentence-level text. Apertus 70B reads each page as numbered sentences and returns people, units, projects, infrastructure, roles and relations, each citing sentence numbers. Code maps the numbers to stored sentence IDs. Names must appear in the cited sentence and roles must be verbatim. Contacts are read by regex, never from model text. A group's expertise is never transferred to its members, and instruction-like page text is withheld (prompt-injection guard). A rule check plus a 70B judge then decides whether the cited sentences really state each claim: **4,314 of 5,130 extracted claims passed, 816 were rejected** (469 by rules, 347 by the judge). Same-name person entities are merged through T08 aliases (A6), which removed 56 duplicate cards. The index is one SQLite file plus vectors, about 138 MB, and it restores in ≈ 7 s.

At query time. BM25 and multilingual-e5-small run in parallel, are max-aggregated per entity *before* weighted reciprocal-rank fusion (BM25 weight 0.25), and are diversified with MMR. News, fact-sheet and navigation pages are not shown as contacts. A deterministic policy labels each answer *confident*, *low confidence*, *too broad* or *no evidence*. In the background, Apertus 70B interprets the request into editable needs, research terms in English and German, and a scope verdict. A second retrieval over those terms produces a refined ranking that never overrides a refusal or a *too broad* outcome. 8B writes per-candidate assessments, which are validated against the candidate's own evidence. If Apertus is slow, 8B takes over interpretation after 12 s; if it is down, the fast results stand.

Pipeline in detail

Stage	What happens	Numbers / settings
Source inventory	Supplied UniBE subdomain inventory + ARTORG + DCB, one YAML per source (domains, paths, languages, budgets)	41 sources; focus sources 200 pages / depth 4, others 50 / depth 2
Crawl	robots.txt (fail closed), ≤ 0.5 req/s per host, back-off, cache; page versions with fetch date and content hash; split into passages and sentences	2,811 pages · 148,404 sentences
Extraction (T07)	Apertus 70B reads numbered sentences → people, units, projects, infrastructure, services, roles, expertise, relations; JSON-schema decoding; injection-like text withheld	5,130 claims · ≈ 1.05M in / 0.39M out tokens · ≈ 30 min
Support check + resolution (T08)	Rules + 70B judge keep a claim only if the cited sentences state it; contacts only via regex and only for current people; same-name persons merged (A6); "former"/"emeritus" read from the page (C3)	4,314 supported · 816 rejected · 105 person merges · 20 marked former
Index	SQLite FTS5 (BM25) + multilingual-e5-small int8 ONNX vectors per passage and per entity profile; incremental rebuild reuses vectors of unchanged text	2,488 served entities · 22,124 vectors · ≈ 138 MB · restore ≈ 7 s
Instant retrieval	BM25 ∥ dense; max per entity before weighted RRF (BM25 weight 0.25); MMR diversity; listing pages demoted ×0.5; news/navigation pages never shown as contacts; duplicates folded	server median ≈ 0.26 s · p95 ≈ 0.5 s · 0 LLM calls
Answer policy (T31)	Deterministic label: confident / low confidence / too broad (with narrowing suggestions) / no evidence (dense < 0.83)	no card without a stored evidence sentence
Apertus refinement (T18/T30)	SSE stream after the first cards: 70B interpretation (needs, EN/DE terms tied to a source word, scope check) → second retrieval over the terms → refined ranking that never overrides *no evidence* / *too broad*	70B deadline 12 s → 8B fallback · 60-min cache · nonsense refused (0/29 real requests refused)
Assessment (T19)	8B labels each candidate direct / related / insufficient with per-need coverage, citing only the candidate's own sentences; parallel batches of 2 with a straggler deadline	≈ 6 s per batch · labelled as AI · never reorders
Shortlist and drafts (T38/T23/T24/T28)	Shortlist with notes and source-only PDF; Apertus brief, e-mail draft and quote translation, withheld if they add contacts or numbers not in the evidence	no recipient, nothing sent · translations labelled "cite the original"

3. Purposeful AI: which model does what, and why

Step	Model	Measured	Decision
Fact extraction (offline, cached)	Apertus 70B	≈ 97% claim precision and 24 persons vs ≈ 48% and 0 for 8B (11 pages, hand-checked); ≈ 30 min for the corpus	70B: quality matters, runs once
Request interpretation and expansion	Apertus 70B	Recall@10 0.65 → **0.82** (frozen evaluation set, in-process); median 4.9 s, background	70B
Same with 8B	Apertus 8B	0.65 → 0.65 (no gain), 1.3 s	Fallback only (12 s deadline)
Candidate assessment	Apertus 8B	≈ 6 s vs 18 s for 70B, similar on clear cases; 86% of expected answers labelled direct/related	8B: speed
Scope check (refuse nonsense)	Apertus 70B	Development: 0/29 real requests refused, 9/10 nonsense probes refused; evaluation check: 0/19 refused	On; 8B verdicts never refuse
Fast-path embeddings	multilingual-e5-small (ONNX int8)	server p95 ≈ 0.5 s, CPU only	Kept; Swiss embeddings on roadmap

All structured output uses the gateway's JSON-schema constrained decoding and is re-validated with Pydantic. Running on the shared Keera gateway taught us concrete rules: flat schemas and short string labels avoid the decoder failures we saw with nested lists.

4. Technical rigour: evaluation

Protocol. The gold set has 36 real partner-search questions grouped by intent: professional, lay, synonym, misleading, partial and absent, in EN/DE/FR/IT/RU. Expected entities are keyed by source URL. Intent groups are split 17 development / 19 evaluation. The set was frozen (gold 0.3, hash `340c4b0a…`) before any evaluation-split run. **Every threshold and weight was chosen on development; each evaluation configuration was scored once.** One evaluation intent viewed before a re-score is disclosed and was not used for tuning.

Frozen evaluation split, Recall@10 (n = 17 scored)	Value
BM25 only (in-process)	0.29
Dense only (in-process)	0.59
Hybrid (in-process, arm64)	0.65
First live version (18 Oct)	0.47
Live today, instant (19 Oct)	**0.77**
Live today, after Apertus refinement	**0.82**
Development split, live, both modes	0.82

Quality and grounding	Value
Cards with a source link	100%
Quote mismatches vs stored sentence (350 cards, 2,754 quotes)	0
Red-team rule violations (18 adversarial queries)	0
Correct abstention on 2 "no answer" questions	0 / 2
Sources represented in served cards	34 / 41
Share of cards from top 3 hosts	40%

The live instant number (x86, 0.77) and the in-process hybrid number (arm64, 0.65) are the same configuration. The difference is two edge-of-list queries (I11-ru, I18-fr) whose int8 embedding scores differ between CPU types (#147). The expansion gain (+0.17) was measured within one setup. The end-to-end mode scores the Apertus-refined list when the server sends one: 14 of 19 evaluation queries; the rest kept the instant list through the guard. n = 17 is small, so we read all numbers as indicative.

Support of evidence (semantic). Quotes are verbatim by construction, but a verbatim quote can still be off-topic. Three samples of 40 cards each were rated *supported* / *partly* / *unsupported*:

- The baseline cards, AI pre-rated and every rating reviewed by Ruihan: 5% / 12.5% / 82.5%. In 28 of 33 unsupported cases the shown sentence was a heading or fragment.
- The current cards rated by two independent model judges: 15% / 28% / 57% and 7.5% / 50% / 42.5%, agreeing on 31 of 40.

Strong matches sit at ranks 1–3, and ranks 4–10 are mostly weak. This is our main quality gap (section 8). An assessment sample (10 live, stratified) had the category right in 6 of 10, with no invented contacts or numbers.

Tried, measured, not shipped.

Idea	Result	Decision
Open cross-encoder rerank of the top 30	arm64: dev 0.82 → 0.88, eval 0.65 → 0.71. **Deployed x86: dev 0.82 → 0.82, +1.2 s per search**	Not shipped
Cut cards far below the best one	dev unchanged with 4.4 instead of 9.8 cards; **eval 0.77 → 0.65**	Flag only (`weaker_match`)
Larger crawl E1 (+1,529 pages, +54%) with a stricter extraction prompt	dev recall unchanged; cards with fields 26% → 20%; supported claims 4,289 → 3,710	Not shipped
Type hint, corpus acronyms, title boost, listing classes	dev unchanged	Off
Relevance-based evidence for assessment	direct/related on expected answers 0.82 → 0.71	Opt-in only

5. Value, cost and scalability

- **Latency.**
  - First results: server median ≈ 0.26 s, p95 ≈ 0.5 s.
  - 10 concurrent users: p95 ≈ 3 s (load test, 18 Oct).
  - Apertus refinement and assessments: median ≈ 14 s, p95 ≈ 17 s after the B1 latency fixes (parallel batches, interpretation cache, straggler deadline), down from 33.5 s / 50.5 s. Measured end to end on the live Space over the development split: median 17.5 s.
  - The analyst never waits for the model: results are shown first, and the refined ranking is offered when it arrives.
- **Tokens.**
  - 0 per plain search.
  - ≈ 1.7k input + 1.2k output per refined request.
  - Processing the whole corpus: ≈ 1.05M input + 0.39M output tokens, run once and cached. Re-runs touch only changed pages.
- **Hosting dominates cost.** The demo Space (CPU Upgrade) costs ≈ USD 22 per month. Keera token prices are not public to us, so we report tokens and formulas (`docs/costs.md`).
- **Scaling.**
  - The fast path is CPU-only and stateless per request. Scaling means replicas over one read-only index.
  - The index rebuild is incremental: unchanged text reuses its vectors. The person merge re-encoded 31 entries in 41 s.
  - Apertus calls are bounded per request: at most 8 candidates, concurrency 4.

6. Sovereign deployability

- **Swiss model on Swiss compute.** Apertus (Swiss AI Initiative) is served through the organisers' Keera gateway on CSCS. The product uses no other LLM.
- **Open, replaceable components.** FastAPI, SQLite FTS5, ONNX Runtime, multilingual-e5-small (MIT) and Apertus (Apache-2.0 per its model card). The LLM endpoint is a single OpenAI-compatible base URL, so a local llama.cpp server can replace the gateway.
- **One command.** `docker compose up` runs the same image as the live Space, restores the index bundle, binds to localhost only and never crawls. A profile adds a local Apertus 8B (community Q4_K_M GGUF, about 7 tokens/s on a laptop CPU).
- **Shown so far:**
  - the fast path on the real index with every outbound socket refused: 0 connection attempts, 255 ms for an unseen German query;
  - an in-container check script that proves egress is blocked.
- **Not shown:** a recorded air-gapped run with the network switched off at OS level, and local Apertus on the query path. This is a declared gap (decision D13).

7. Responsible by design: crawling, data, AI text

Polite crawling

- robots.txt obeyed; if unreadable, the site is not crawled (fail closed); no redirects outside the allowed domains.

Data minimisation, removal, retention

- Only public professional pages from the supplied sources. Contacts are shown only through an official-

- At most **0.5 requests per second per host**, identifying User-Agent, back-off on 408/429/5xx with `Retry-After` honoured.
- Fixed budgets per source (pages, depth, sitemaps, discovered URLs, 10 MB per response); cached pages reused.
- **No fetch at query time:** searches read only the stored corpus.

contact field, never quoted from text.
- Company descriptions are not stored and kept out of logs; logs are kept 30 days.
- Removal on request: suppressed records are skipped by the crawler, filtered at serving time and answered with `410`. Stored copies are purged within 30 days (proposed retention). The contact address is still to be published.
- Evidence older than 365 days is marked stale. Former staff are never shown as current; 20 persons were marked from their own pages.
- AI text is always labelled; e-mail drafts are never sent.

Fair discoverability. Scores are max-aggregated per entity, so a group with one page competes on equal terms with a page-rich institute. Across 350 served cards, 34 of 41 sources appear and no single host dominates (largest 15%). There is deliberately no equal-exposure quota: relevance comes first. **Legal.** Built for the Swiss nDSG and the GDPR where applicable: only public content, data minimisation, removal and retention as above. Copyright of the source pages stays with the institutions; we store sentences only to quote and link them. Outputs are to be published as Open Commons per the challenge terms (§6).

These are design commitments, not a claim of legal compliance.

What we delivered

- **Live service** on a Hugging Face Space, one-command deploy with smoke test and rollback, `docker compose up` runbook and in-container air-gap check.
- **Corpus and knowledge**: 41 sources, 2,811 pages, 4,314 verified claims; an extended crawl (+1,529 pages) collected and measured, not yet shipped.
- **Search**: hybrid multilingual retrieval, answer policy, contacts-only cards, "Why this result" (matched words, channels, ranks).
- **Apertus features**: request interpretation with editable needs, term expansion, scope check, refined ranking, per-candidate assessment, brief, e-mail draft, quote translation.
- **Interface**: EN/DE/FR, instant cards then optional refined ranking, confidence labels, "people to contact" strip, out-of-scope message, shortlist with notes and PDF export.
- **Operations**: protected admin with per-source coverage, refresh executor, feedback persisted across restarts, suppression and removal (410), rate limiting.
- **Evaluation**: frozen gold set and HTTP harness (instant and end-to-end modes), grounding checker, red-team suite, load test, support-rating sheets.
- **Documentation**: README with architecture, crawling safeguards, data policy, decisions D1–D16, measurements, attribution.

8. Implementation feasibility, readiness and limits

How we worked. Every challenge requirement became a GitHub issue with acceptance criteria (175 issues and PRs, 240 commits on `main`). Design choices are recorded as decisions D1–D16. Changes were merged after CI (360+ deterministic tests), ranking changes were merged only after a development-split measurement; evaluation-split runs were disclosed. Ideas that did not measurably help were not shipped (section 4).

Operation. One-command deploy with smoke test and commit check; rollback by restoring the previous bundle. Feedback and admin state survive restarts; rate limiting; protected admin with per-source coverage. CI runs 360+ deterministic tests with no network and no paid inference. A new source is a YAML file validated by the loader (example: BFH engineering research).

Needed for production: Swiss or on-premises hosting; scheduled refresh crawls; monitoring and alerting; single sign-on; a staffed removal contact; a recorded air-gapped acceptance run; partner review of the gold set (it was written by the team).

Limitations.

- Small samples: 17 scored evaluation queries, one location for latency.
- Lay wording without the lab's terms is the weak spot. A juror's live query, "old people care and supply", returned low-confidence cards: the geriatrics clinic and the gerontechnology group exist in the index, but the instant path did not reach them. The 8B fallback (70B was over its 12 s deadline) added only weak terms, and the relevant cards sat under "More results".
- Structured fields cover 26% of cards; every card has quote and link.

- Ranks 4–10 are often weak matches.
- The 2 "no answer" questions still return low-confidence cards.
- The same project can appear twice through its DE and EN pages.
- French and Italian source content is sparse.
- The local Apertus model is an unofficial conversion.

Next.

- Swiss / Apertus embeddings, measured on the same gold set.
- Ship the larger crawl with an extraction prompt that keeps field completeness.
- Show the best-matching sentence first.
- Merge language versions.
- Team shortlists with contact status and CRM export.

9. Dependencies, licences and provenance

Model and runtime: Apertus v1.5 8B / 70B, Swiss AI Initiative (Apache-2.0 per model card; via Keera/CSCS) · multilingual-e5-small, revision 614241f6 (MIT) · ONNX Runtime (MIT) · llama.cpp b11719 (MIT, local probe only).
Application: FastAPI, Pydantic (MIT) · Uvicorn, httpx (BSD-3-Clause) · SQLite (public domain).
Not shipped: cross-encoder mmarco-mMiniLMv2-L12 (Apache-2.0, evaluated only).
Data: public pages of the 41 supplied UniBE, ARTORG and DCB domains, crawled 18–19 Oct 2026. The corpus and index are not published in the repository because they contain names from public pages. The full inventory is in `docs/attribution.md`.

Appendix: results by question type and cost formulas

Frozen evaluation split (live, 19 Oct)	n	Instant	+ Apertus
Cross-lingual (EN/DE/FR/IT/RU)	9	0.67	0.78
Synonym	2	1.00	1.00
Lay wording	2	0.50	0.50
Professional · misleading · partial · person/unit	1 each	1.00	1.00
All scored questions	17	**0.77**	**0.82**
"No answer" questions: correct abstention	2	0/2	0/2

Cost model (Keera prices not public to us, so tokens are reported): corpus processing ≈ 1.05M input + 0.39M output tokens, once, then only changed pages; an enriched request ≈ 1.7k input + 1.2k output tokens; a plain search 0 tokens. Hosting: CPU Space at USD 0.03 / h ≈ USD 22 / month. At an assumed CHF 1 / 3 per 1M input / output tokens for 70B (placeholder, not a quote), 300 enriched requests a year cost under CHF 1 in tokens; hosting dominates.

Reproduce: `python -m evaluation run --base-url <app> --split development [--enriched]`, then `python -m evaluation report`; grounding: `python -m evaluation.grounding`; air-gap check: `docker compose exec -T app python - < scripts/airgap_check.py`.